Roll No. | | | | | | | | | | | | Total No. of Pages: 02

Total No. of Questions: 13

B.Pharmacy (Sem.-2)

BIOCHEMISTRY

Subject Code: BP-203T

M.Code: 74969

Date of Examination: 18-05-2026

Time: 3 Hrs. Max. Marks: 75

INSTRUCTIONS TO CANDIDATES:

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Attempt all questions:**

(a) Define endergonic and exergonic reactions with one example.

(b) Write the biological significance of ATP

(c) What is G6PD deficiency?

(d) Differentiate between glycogenesis and glycogenolysis.

(e) Write a note on oxidative phosphorylation.

(f) Define hyperuricemia.

(g) Write two therapeutic applications of enzymes.

(h) Name any two coenzymes and their functions.

(i) Define the term energy-rich compound with an example.

(j) Difference between essential and non-essential amino acids.

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SECTION-B

2. Explain the structure of DNA and RNA with functions and mechanisms of DNA replication.
3. Discuss β-oxidation of palmitic acid and its significance in energy production.
4. Explain the glycolysis pathway along with its energetics and biological significance.

SECTION-C

5. Write a note on the HMP shunt and its significance.
6. Explain the electron transport chain with its mechanism.
7. Write a short note on ketone body formation.
8. Explain the catabolism of phenylalanine and its metabolic disorders.
9. Discuss enzyme inhibition with suitable examples.
10. Discuss the Krebs cycle. https://www.pyqonline.com
11. Write the synthesis and significance of dopamine.
12. Draw the urea cycle with its significance.
13. Give the biological significance of cholesterol.

NOTE: Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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